CS19541 – COMPUTER NETWORKS LAB

Practical 12 – Echo and Chat Client-Server Using TCP/UDP Sockets

AIM

To develop echo and chat applications between a client and a server using TCP or UDP sockets.

PART A – ECHO CLIENT-SERVER: ALGORITHM

1. Create a server socket and bind it to a selected IP address and port.
2. For TCP, listen for a connection and accept the client.
3. Let the client send a text message to the server.
4. Receive the message at the server and send the same text back.
5. Display the returned message at the client.

PART A – INPUT / OUTPUT

Input: A text message typed by the client.
Output: The server sends the same message back, and the client displays it.

PART B – CHAT CLIENT-SERVER: ALGORITHM

1. Create and configure the client and server sockets.
2. Connect the endpoints using TCP, or exchange datagrams with `sendto()` and `recvfrom()` for UDP.
3. Enable the client and server to send and receive messages.
4. Print each message at the receiving endpoint.
5. End the session when a participant enters `exit`.

PART B – INPUT / OUTPUT

Input: Messages typed at the client and server terminals.
Output: Both sides display the messages exchanged until the chat is closed.

PROCEDURE

1. Save the server and client code in separate files, or use a program that allows the role to be selected.
2. Start the server program first.
3. Launch the client in another terminal on the same computer or on a reachable machine.
4. Send sample text and check the echo or chat response.
5. Type `exit` to stop the chat session.

OBSERVATION

The echo application returned the client's message. The chat application exchanged text between the client and server through socket communication.

RESULT

The echo and chat client-server applications were implemented and tested using TCP/UDP socket programming.